
WS PROJECT 2

MUSIC TRENDS

ÂNGELA RIBEIRO (109061) | ANTÓNIO ALBERTO (114622) | HUGO CASTRO (113889)

ONTOLOGY · INFERENCE · LINKED DATA — JUNE 2026



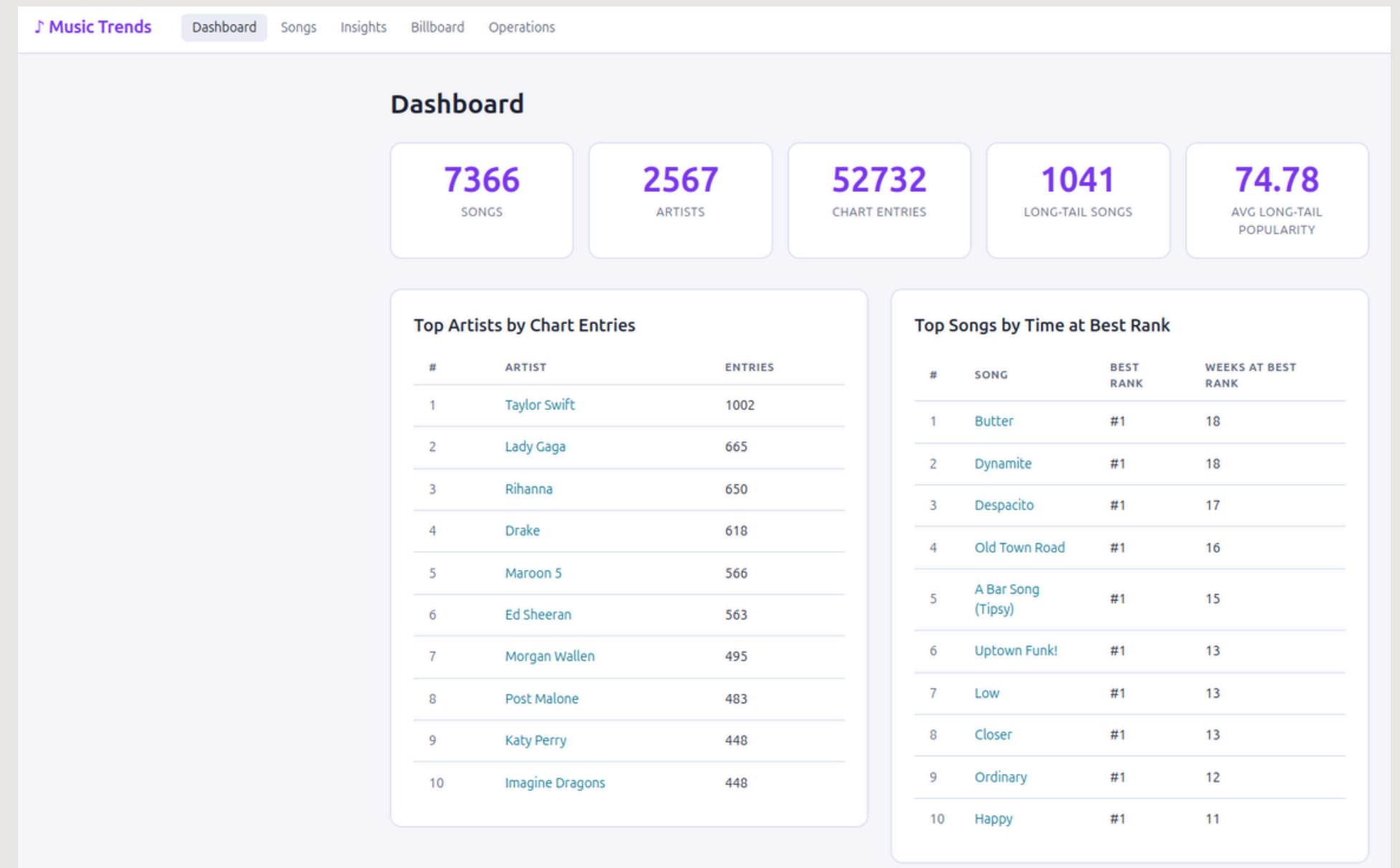
— FROM PART 1 TO PART 2

Part 1 — the foundation

- Kaggle datasets converted to RDF
- Stored & queried in GraphDB / SPARQL
- Django app to explore the data

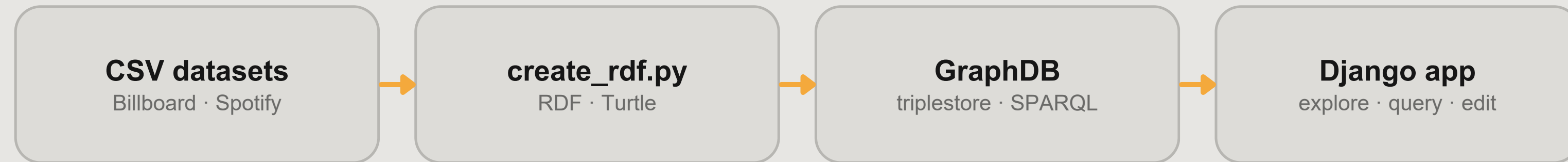
Part 2 — adding semantics

- An OWL/RDFS ontology of the domain
- SPIN rules: classification + new links
- Enrichment from DBpedia & Wikidata
- Semantics published in pages (RDFa & Microformats)

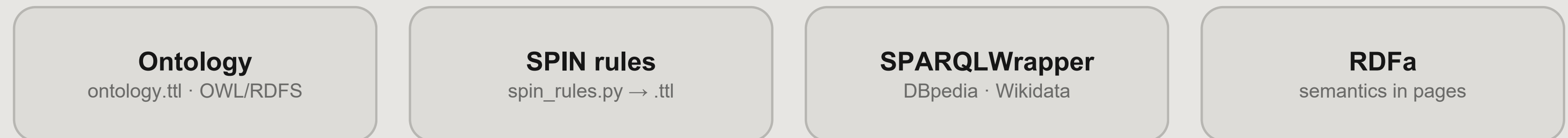




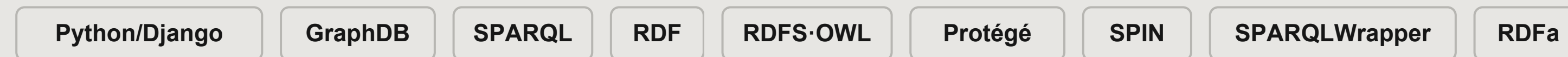
— ARCHITECTURE & STACK



Semantic layer wired into the pipeline



Stack



— ONTOLOGY (RDFS / OWL)

- **Hierarchy** subclasses go beyond the raw data
- **Beyond data** Country · RecordLabel · TimePeriod
- **inverseOf** performer ↔ performed
- **subPropertyOf** mainArtist, featuredArtist ⊆ performer
- **equivalent / symmetric** song ≡ entrySong · collaboratedWith
- **owl:sameAs** links artists to Wikidata / DBpedia
- **Annotations + Protégé** labels/comments · HermiT reasoner

```
owl:Thing
├ Song
│   ├── ChartedSong ← inferred
│   │   ├── HitSong · LongTailSong ← inf.
│   └ Artist
│       ├── SoloArtist · BandOrGroup
│       ├── HitArtist · TrendingArtist ← inf.
│   └ ChartEntry ⊆ prov:Entity
├ TimePeriod ⊃ Decade · Season
└ Country · Genre · RecordLabel · Album
```



— SPIN INFERENCE RULES

Independent Python module (spin_rules.py) — rules RDFS/OWL can't derive alone (need FILTER / dates / aggregation).

Rule	Produces	Condition
ChartedSongRule	type:ChartedSong	song in ≥ 1 chart entry
HitSongRule	type:HitSong	entry with rank ≤ 10
LongTailSongRule	type:LongTailSong	max weeks on chart ≥ 20
HitArtistRule	type:HitArtist	performs a HitSong
TrendingArtistRule	type:TrendingArtist	top-10 entry since 2024
AppearsInChartRule	pred:appearsInChart	artist $\rightarrow$ chart (new link)
CollaboratedWithRule	pred:collaboratedWith	shared song (symmetric link)

Exported as SPIN (sp:/spin:) $\rightarrow$ spin_rules.ttl · materialized on GraphDB via INSERT ... WHERE



— LINKED DATA

Enrichment — DBpedia & Wikidata

- **SPARQLWrapper** both endpoints, programmatically
- **Wikidata** country, genre, image, dates, MBID
- **DBpedia** thumbnail, birth place, abstract bio
- **owl:sameAs** artists linked to external entities
- **Coverage** $\approx$ 1903 / 2565 artists enriched

← Songs Explorer

Avicii

edm pop dance pop pop dance pop music house music electronic dance music



About

Swedish electronic music producer (1989–2018)

COUNTRY	FROM	BORN
Sweden	Stockholm	1989-09-08

[Official site ↗](#) [Wikidata ↗](#) [DBpedia ↗](#) [MusicBrainz ↗](#)

Source: DBpedia & Wikidata



— RDFa - ARTIST DETAILS PAGE

Ava Max

pop

dance-pop



About

American singer and songwriter (born 1994)

Text source: Wikidata description (fallback).

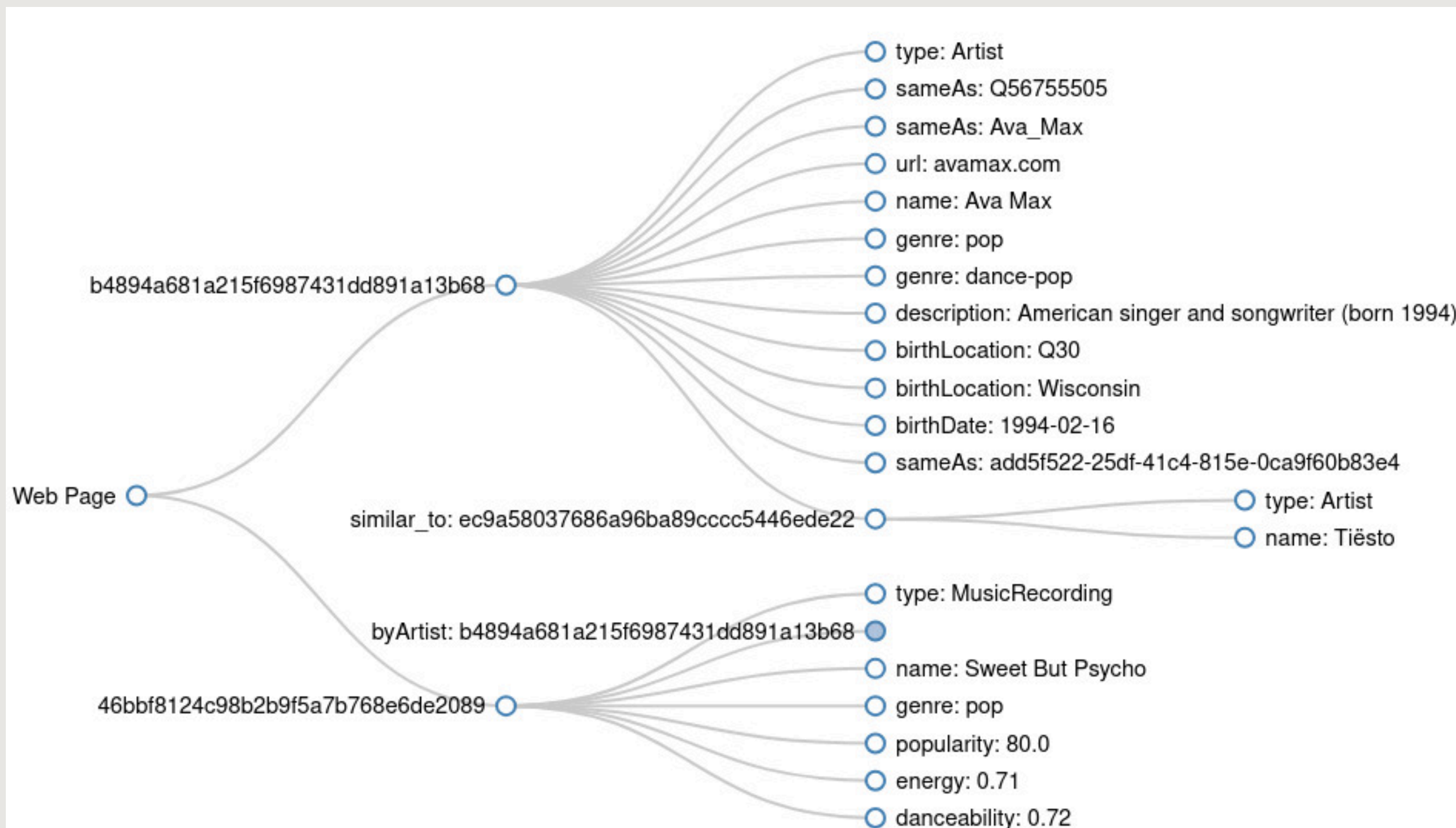
COUNTRY	FROM	BORN
United States	Wisconsin	1994-02-16

[Official site ↗](#) [Wikidata ↗](#) [DBpedia ↗](#) [MusicBrainz ↗](#)

Structured data: Wikidata. Biographical text: DBpedia.



— RDFa - Example





— MICROFORMATS - SONG DETAILS PAGE

Die With A Smile

Show Lyrics

by Lady Gaga

Details

Genre

pop

Featured

Bruno Mars

Audio Features

No audio features available.



— MICROFORMATS

```
{
  "items": [
    {
      "type": ["h-entry"],
      "properties": {
        "name": ["Die With A Smile"],
        "url": ["http://music.org/song/4d307eb0a98e58d0d1dc2492f76638de"],
        "category": ["pop"],
        "author": [
          {
            "type": ["h-card"],
            "properties": {
              "name": ["Lady Gaga"],
              "url": ["http://music.org/artist/f4a1462da44612fd2c5e1adc4fb7842c"]
            }
          },
          {
            "type": ["h-card"],
            "properties": {
              "name": ["Bruno Mars"],
              "url": ["http://music.org/artist/ce178f34069e29a506c407052db01ca7"]
            }
          }
        ]
      }
    }
  ]
}
```



PROTÉGÉ

MusicOntology (http://music.org/MusicOntology) : [/home/angela/Desktop/4º ano/2º Semestre/WS/projeto/Music-Trends-W...

File Edit View Reasoner Tools Refactor Window Help

< > Music Trends Ontology (http://music.org/MusicOntology)

Active ontology x Entities x Individuals by class x DL Query x

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: [Icons] Inferred

- owl:Thing
 - Album
 - Artist
 - Band or Group
 - Hit Artist
 - Solo Artist
 - Trending Artist
 - Chart
 - Country
 - Genre
 - prov:Entity
 - Chart Entry
 - Record Label
 - Song
 - Charted Song
 - Hit Song
 - Long-tail Song
 - Time Period
 - Decade
 - Season

Nothing Selected

Git: dev Reasoner active ☒ Show Inferences

MusicOntology (http://music.org/MusicOntology) : [/home/angela/Desktop/4º ano/2º Semestre/WS/projeto/Music-Trends-W...

File Edit View Reasoner Tools Refactor Window Help

< > Music Trends Ontology (http://music.org/MusicOntology)

> Song > Charted Song > Long-tail Song

Active ontology x Entities x Individuals by class x DL Query x

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: Long-tail Song [Icons] Inferred

- owl:Thing
 - Album
 - Artist
 - Band or Group
 - Hit Artist
 - Solo Artist
 - Trending Artist
 - Chart
 - Country
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 - prov:Entity
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 - Record Label
 - Song
 - Charted Song
 - Hit Song
 - Long-tail Song
 - Time Period
 - Decade
 - Season

Long-tail Song — http://music.org/type/LongTailSong

Annotations Usage

Annotations: Long-tail Song [Icons]

Annotations +

- rdfs:label** [language: en]
Long-tail Song
- rdfs:comment** [language: en]
Song with long persistence on charts (high max weeks on chart).

Description: Long-tail Song [Icons]

Equivalent To +

SubClass Of +

- 'Charted Song'

General class axioms +

SubClass Of (Anonymous Ancestor)

Instances +

Target for Key +

Disjoint With +

Disjoint Union Of +

Git: dev Reasoner active ☒ Show Inferences



NEW FEATURES & INSIGHTS

Enriched artist detail page

Artist pages now show external data (Wikidata/DBpedia), with fallback behavior when fields are missing.

Long-tail semantic inference

LongTailSong is now inferred through a SPIN rule (based on chart longevity), instead of coming directly from the raw dataset.

Lyrics viewer

On-demand lyrics via the Irc.lib.net API

Country drill-down

Top artists & songs by enriched origin country

Partial semantic publishing in HTML

RDFa and microformats were added to key pages (partial/prototype coverage, not full-site coverage yet).

New dashboard long-tail metrics

The dashboard now includes long-tail indicators and rankings (count, top songs, top artists, average popularity).



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MUSIC TRENDS

LIVE
DEMO



— CONCLUSIONS

- **A semantic system** ontology + inference + linked data
- **Inference adds knowledge** auto-classification & new relations
- **Web of data** DBpedia & Wikidata enrich the dataset
- **Meaning in the pages** RDFa makes the UI machine-readable

Deliverables

- facts_only.ttl — facts, no typology
- ontology.ttl — ontology-only layer
- integration_protege.ttl — for Protégé
- spin_rules.py / .ttl — 7 rules
- setup.sh full — one-command run

THANK YOU — QUESTIONS?



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